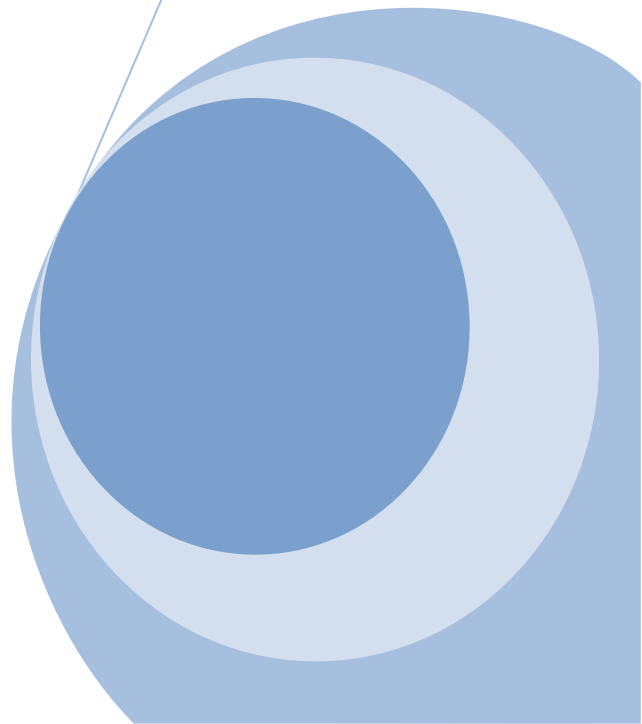




PART-2

FUNDAMENTAL OF JAVA

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Basics fundamental of java

1. Character Sets
2. Keyword/Reserved Words
3. Identifier/User Defined word
4. Data types
5. Variables
6. Constants
7. Literals
8. Operators
9. Control Statements
10. Arrays

1. Character Sets

Alphabets	A – Z/a – z
Digits	0-9
Special Characters	+ - * % @ & \$! < > { } = ~ ^ etc.
While Space Character	New line (\n), Carriage Return (\r), Tab (\t), Backspace(\b) etc

2. Keyword/Reserved Words

- ❖ Keywords are simple English word which are having predefine meaning in any (like C, C++, Java etc) programming language.
- ❖ Keywords are reserved word that provided by vendor. Meaning of it can't be modified.
- ❖ All the keyword defined in Lower Case.
- ❖ We can't use keywords names for variables, methods, classes, or as any other identifiers.
- ❖ **true**, **false** and **null** is not a keyword, it reserved word.
- ❖ There are **49** keyword is defined up to java 2. **enum** (50th) is new keyword add in java5

Data type (8)	boolean int	byte long	char float	short double
Class & Object (9)	class implements instanceof	interface this	enum super	extends new
Access Modifier (3)	private	protected	public	
Modifier (9)	final transient strictfp	native volatile	abstract static	synchronized const *
Package (2)	package	import		
Control Statements(12)	if default for	else do continue	switch while return	case break goto *
Exception Handling (6)	try throws	catch assert	finally	throw
Other Data type (1)	void			

- ❖ There is no implementation available for Const & goto in java. you can't use these keyword in Java Program.

3. Identifier/User defined words

- ❖ Identifiers are the names those will be used to identify the programming elements like Classes, Methods and Variable etc uniquely.
- ❖ Identifiers are also known as User Defined words.

Rules to follow when you define Identifiers:-

- ⇒ Identifiers can contain alphabets, Digits, and two special Symbol i.e Dollar (\$) and Underscore (_).
- ⇒ First Character of an identifier must be an Alphabet or Dollar(\$) or Underscore (_)
- ⇒ You cannot use Digit as first char of Identifiers.
- ⇒ Keywords or Reserved words can't be used as Identifiers.
- ⇒ There are no space between two word and also cannot use (-) between two words.

Example:-

Identifiers	valid/Invalid	Identifiers	valid/Invalid
Hello	v	Int	v
getClassName	v	int	inv
1stclass	inv	true	inv
student name	inv	student-email	inv
Studentname	v	student_email	v
\$pay	v	\$500	v
pay100rs	v	_123	v
Null	inv	_name	v
null	v	num*5	inv

4. Data types

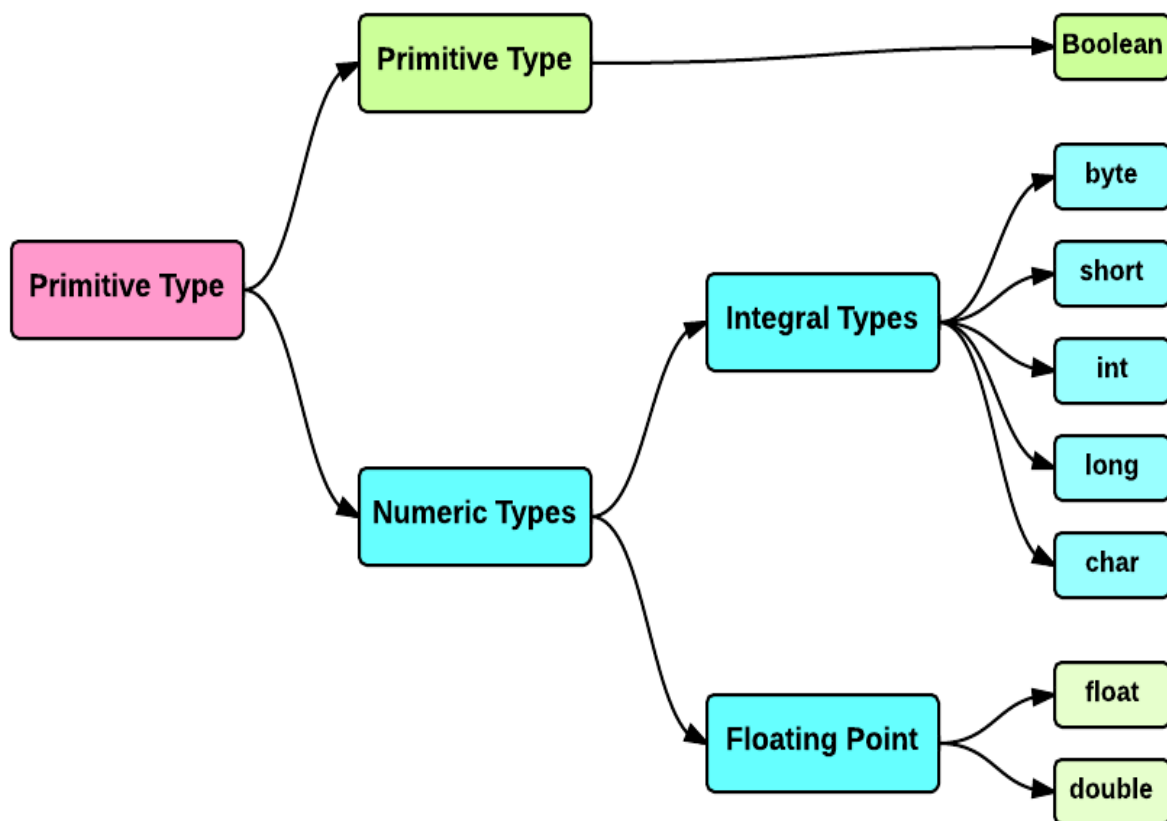
- ❖ Data type specifies the size and type of values that can be stored in an identifier.
- ❖ Here, size meaning is Amount of memory allocation required for your data.

There are two type of data type:-

1. Primitive Data type.
2. User Defined Data type.

1. Primitive Data type

- ❖ There are 8 Primitive data types available in java.
- ❖ There are following primitive data types
 1. Boolean
 2. byte
 3. char
 4. short
 5. int
 6. long
 7. float
 8. double



2. User Defined Data type

- ❖ There are four type of user define data types:
 - ⇒ class type
 - ⇒ Interface type
 - ⇒ Enum type (java 5)
 - ⇒ Annotation type (java 5)

Type	Size		Default value	Min value	Max value
	Byte	Bits			
boolean	N/D	N/D	false	true or false	
byte	1	8	0	-2^7 (-128)	2^7 -1(127)
char	2	16	ASCII-0	0	2^{16} -1(65535)
short	2	16	0	-2^{15} (-32768)	2^{15} (-32767)
int	4	32	0	-2^{31} (-2147483648)	2^{31} -1(-2147483648)
long	8	64	0	-2^{63}	2^{63} -1
float	4	32	0.0	1.4E - 45	3.40E 38
double	8	64	0.0	4.9E - 324	1.79E308
Reference /User define data type	8	64	null	Refer of the corresponding type Object	